Automated Dual-Channel Fluorescence Confocal Image Analysis Workflow for Fiji/ImageJ (1.54p) - MChimanal

Last updated: Aug 13, 2026 (*coding in UTF-8*),

Readme updated: Aug 13, 2026

Software version: 1.0.0

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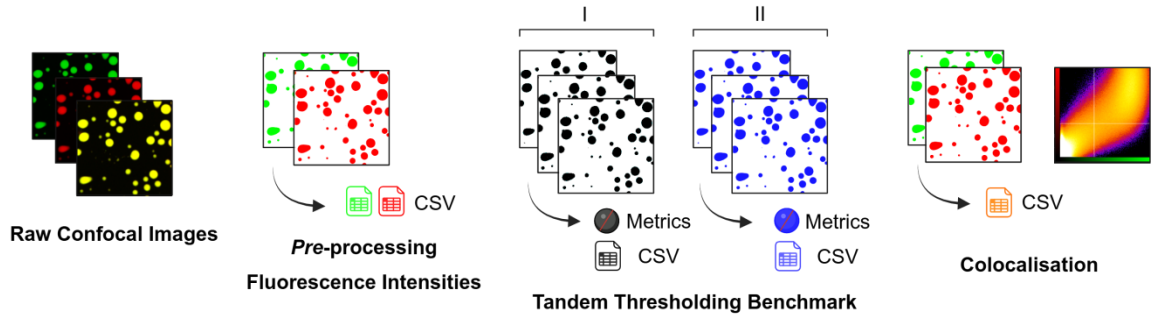

This project is an end-to-end, Python and Fiji pipeline for dual-channel fluorescence microscopy (*.lsm*) images. It auto-normalises channels, tests multiple ImageJ threshold models (doi.10.1117/1.1631315), extracts morphology metrics (droplets number, area occupied, diameter), plots fluorescence intensity for both of the channels and summary visuals, runs *JACoP colocalisation* (doi:10.1111/j.1365-2818.2006.01706.x), and exports everything to clean outputs with organised file housekeeping.

Who it's for?

Scientists and imaging specialists who need reproducible, high-throughput quantification of two-colour fluorescence data-especially in biomolecular condensate studies-and computational biologists who want ready-to-analyse tables and figures without babysitting ImageJ/Fiji.

Overview

This single-file Python workflow automates end-to-end analysis of two-channel Zeiss *.lsm* confocal images,

1. **Fluorescence intensities.** The pipeline first determines the optimal min/max intensity range on the reference channel (**C2** by default) and applies the same limits to channel **C1**, ensuring both channels share an identical dynamic range. This prevents misleading enhancement of dimmer channels due to automatic brightness and contrast adjustments, which would otherwise stretch intensity based on background pixels. Fluorescence intensity is quantified by multiplying each grey-level value by its pixel count and summing the resulting contributions. The final value represents the total fluorescence from all signal-contributing objects visible in the image. Visual scaling is harmonised across channels,
2. **Thresholding benchmark**. The workflow supports evaluation of any number of Fiji's (*19*) built-in thresholding models (from just one to all), depending on user selection. Each model is applied to the intensity-adjusted **C2** channel, which acts as the input

mask. To separate merged droplets or overlapping regions, a watershed transform is automatically applied after thresholding and conversion to a binary mask.

For each model, the pipeline generates:

- the original input image and a merged overlay of channels **C1** and **C2**,
- binary masks resulting from each thresholding model,
- detailed measurement tables

Two parallel analyses are then performed for each model:

- one excluding objects on the edges of the image,
- one including objects on the edges of the image.

For each object, the following metrics are calculated:

- area
- perimeter
- two independent estimates of diameter (one from area, one from perimeter)
- object count

All measurements are saved as **CSV** files, separately for edge-included and edge-excluded analyses. Additionally, all per-model and per-metric statistics are merged and organised into structured **Excel** files.

Metric extraction. These statistical summaries include the median, first quartile (Q_1) and third quartile (Q_3) for every threshold model.

Model rankings. To identify the most reliable thresholding, the pipeline uses a Shewhart-style control chart approach:

- Each metric (Area, diameter, counts) it computes the median across all selected models for two parallel analyses,
- Classify threshold models according to whether all morphology metrics fall within ±1 SD, ±2 SD, ±3 SD, or outside ±3 SD of the corresponding overall medians.

All masks, overlays, and statistical summaries are saved, both individually and as part of a consolidated dataset. A final comparison plot summarises the thresholds' performance side-by-side, enabling both automatic model selection and manual review.

3. **Colocalisation.** The pipeline uses the BIOP JaCoP plugin to compute multiple colocalisation metrics between channels C1 and C2, including:
    - Pearson, Spearman, Manders, Li's ICQ, Costes, and overlap coefficient,
    - Intensity scatter plots.

    To minimise visual artefacts, a primary high-resolution macro variant is executed first. If the resulting fluorogram is missing or exceeds the predefined gridiness threshold, an alternative compressed variant is executed and the better result is retained:

    - a high-resolution version,

- a compressed version (with binning and no stroke outlines).

A custom “gridiness” score is used to evaluate scatter plots quality (based on variance across plot bins). The pipeline then:

- parses the ImageJ console log to extract all key metrics,
- selects the better macro run based on gridiness,
- saves clean CSV and Excel tables with results,
- exports the final scatter plot image.

Output

For every (*.lsm*) image the pipeline spins up a dedicated results folder whose name is a cleaned-up version of the original filename. At the top level you'll find quick-look PNGs (C1, C2, merged (C1 and C2), threshold grid, diameter and droplets number scatter plot) plus two workbooks that collect all object statistics, Shewhart scores and colocalisation metrics. Raw data are stored as CSVs.

<working-folder>/

├─ **Channel previews and histograms**

│ ├─ <base>_C1_original**.png** # 8-bit, LUT-applied preview (channel 1)

│ ├─ <base>_C2_original**.png** # 8-bit, LUT-applied preview (channel 2, reference)

│ ├─ intensity_distribution_histograms**.png**

│ ├─ fluorescence-intensities-C1**.xlsx**

│ └─ fluorescence-intensities-C2**.xlsx**

├─ **Thresholding overview**

│ ├─ grid_threshold_summary**.png** # one sheet with all masks side-by-side

│ ├─ statistics**.xlsx** # per-model medians with Q_1 and Q_3

│ ├─ statistics_edges**.xlsx** # the same, but "objects on edges" only

│ ├─ statistics_plots**.png** # 3-panel plot: area, diameter, counts

│ ├─ summary_threshold_results**.xlsx** # full per-object tables

│ ├─ summary_threshold_edges_results**.xlsx**

│ └─ threshold_model_predictions**.xlsx** # ±1 SD / ±2 SD Shewhart ranking

├─ **Colocalisation**

│ ├─ <base>_Merge**.png** # C2 + C1 channels

│ ├─ <base>_Fluorogram**.png** # chosen high-quality scatter-plot

│ ├─ <base>_colocalization-table**.csv**

│ ├─ <base>_colocalization-table**.xlsx**

│ └─ <base>_colocalization-results**.txt** # raw JaCoP console log with results

├─ **edges analysis/** # Overlays for thresholding including the edges

│ └─ <base>_edges_<Model>**.png** (× N)

├─ **edges csv**/ # CSV tables for the same edge objects

│ └─ edges_<Model>**.csv** (× N)

├─ **csvs**/

│ ├─ results_<Model>**.csv** (× N)

│ ├─ C1-intensity-histogram**.csv**

│ ├─ C2-intensity-histogram**.csv**

│ ├─ <base>_colocalization-table_VAR0**.csv**

│ └─ … other auxiliary CSVs

└─ **macros**/ # All ImageJ/Fiji macros for full provenance

├─ macro_analysis**.ijm**

├─ macro_coloc_VAR0**.ijm**

└─ macro_coloc_VAR1**.ijm**

Hardware and System Requirements

The workflow has been tested on Microsoft Windows 10-11, Ubuntu Linux 25.10, and macOS 26.6.1 using Python 3.11.9 and Fiji/ImageJ 1.54p. No specialised hardware or GPU is required. A standard desktop computer with ≥8 GB RAM is sufficient.

Installation

Typical installation takes approximately 15-30 min on a standard desktop computer, excluding download time for Fiji and Anaconda.

Setting up Fiji

1. **Install Fiji. Grab the latest bundled Fiji.app** (*v. 1.54p* at the time of publishing) from (https://fiji.sc/). Once installed, run the updater and add the PTBIOP plugin if you plan to run *JaCoP colocalisation* on your two-channel images,
2. **Configure Fiji for condensate analysis**,
    - Open original image. In *Set Measurements* tick the following: Area, Std Dev, Min & Max, Area Fraction and Perimeter,
    - *In Analyze* → Particles tick the following: Display Results, Clear Results, Summarize, Add to Manager and Overlay.
    - *In Plugins* → *BIOP*→ Image Analysis → *JaCoP* setup channel A as 1, channel B as 2 and tick the following colocalisation result options: get Pearsons Correlation, Get SpearmanRank Correlation, Get Manders Coefficients, Get Overlap Coefficients, Get li ICA, Get Fluorogram and Set Advanced Parameters

Setting up Python

1. **Install Anaconda / Miniconda**. Download the latest installer from and follow the on-screen instructions for your OS (https://www.anaconda.com/)
2. **Match the project's Python version.** The scripts were developed under Python *3.11.9*. Python 3.11.9 is the tested and recommended version.
    - Open an Anaconda terminal and create a 3.11 environment

    ```
    conda install python=3.11.9
    ```

3. **Update conda in terminal,**

    ```
    conda update conda
    ```

4. **Install required packages,**

    ```
    conda install –c conda-forge opencv py-opencv pandas pillow matplotlib seaborn scipy openpyxl
    ```

5. **Locate the user configuration section near the beginning of the script and edit.** In version 1.0.0, these settings are located around lines: 232, 1024 and 1045 with the direct path to Fiji.app**,**

    ```
    Windows: folder_path = "C:/Users/username/fiji-win64/Fiji.app/ImageJ-win64.exe"

    Ubuntu/Linux: folder_path = "/home/username/Fiji.app/ImageJ-linux64"

    macOS: folder_path = "/Applications/Fiji.app/Contents/MacOS/ImageJ-macosx"
    ```

6. **Set your working directory.** In version 1.0.0, update the line 1103 to the directory that contains your .lsm images. Only files whose names start with exp are processed, so make sure your images follow that convention,

```
Windows: folder_path ="C:/Users/username/"

Ubuntu/Linux: folder_path ="/home/username/"

macOS: folder_path ="/Users/username/"
```

7. **Choose which threshold algorithms to benchmark**. On line 1104 edit the list called *threshold_models*. Provide any subset of Fiji's 19 built-in methods e.g. "Otsu", "Li", "Moments",
8. **Tidy filenames** *(optional)***.** If your raw image names carry distracting tags pH, salt, time points, etc. add those fragments to the *shorten_file_name* list on line 41. The helper function will strip them automatically so all outputs stay clean and readable.

How to run

Run the script from the terminal or Spyder program.

```
python github_machine_macro-multi_biomolecular-condensates.py
```

All *.lsm* images in *folder_path* with names that start with *exp* will be processed, and a results folder will be created for each image.

Typical Runtime

Processing one 1024 × 1024 two-channel .lsm image using all 19 thresholding algorithms takes approximately 1-3 minutes on a standard desktop computer (16 GB RAM, Windows 11).

Demo

Download and extract the example_data and example_output directories. The example datasets are provided as split ZIP archives. Download all .z01, .z02, ... and .zip parts into the same directory before extraction. Set folder_path to the example_data directory and run:

```
python github_machine_macro-multi_biomolecular-condensates.py
```

Reference demo runtime: approximately 2 min on an Intel i7 desktop computer with 64 GB RAM. The generated numerical outputs should reproduce the reference results deposited in example_output.

Troubleshooting

On systems where Windows uses the *CP-1250* (Central European) code page, *UTF-8* decoding of Fiji's console output may fail on greek characters e.g. µ etc.

In that case, decode with *encoding="cp1250"* (you can run *chcp* to check the active code page 65001 = UTF-8, 1250 = CP-1250) and keep errors="*replace*" to prevent crashes.

Archived Versions

Software version 1.0.0 is permanently archived in RepOD: https://doi.org/10.18150/2K3PB9

Acknowledgments

This script and pipeline were designed and written by Michal Kacper Bialobrzewski (https://github.com/bialobrezuski). Please cite the following paper when using or adapting this code: https://doi.org/10.64898/2026.07.16.738963. I would like to acknowledge Zuzanna Staszalek, Barbara Klepka, Weronika Chmura, Zuzanna Hirsz, and Marlena Stal for their valuable contribution in checking the workflow of the code.